Training Leaves Traces in Residual Branch Products for Model Lineage Verification

Anonymous submission

Abstract

Open-weight model checkpoints are routinely fine-tuned, quantized, pruned, merged, and redistributed without reliable lineage metadata. Existing provenance tools are brittle in this setting: hashes fail after any weight change, metadata requires trust, and watermarks must be inserted before release. We study post-hoc white-box weight-lineage verification: given a reference checkpoint and a suspect checkpoint with compatible residual architecture, can we determine whether the suspect inherits parameters from the reference after common checkpoint transformations? This is distinct from behavioral similarity: under weight-level lineage, distilled students trained from fresh weights are non-descendants, even when they imitate a teacher's outputs.

We show that residual-network training induces an intrinsic fingerprint in residual-branch products. Correctly paired branch products develop diagonal-dominant fingerprint, measured by a trace/Frobenius score, that is absent at initialization, disappears under incorrect factorization, and does not appear in non-residual controls. We use this signal to align blocks between checkpoints and compute a model-level lineage score from centered residual-branch signatures. Across GPT-2, BERT, LLaMA-2, Mistral, Qwen2.5, DeepSeek-R1, ViT, Whisper, and ResNets, the fingerprint recovers block correspondences with 91–100% accuracy. On controlled lineage benchmarks, this model-level lineage score yields complete separation: descendants produced by fine-tuning, pruning, quantization, and LoRA merging score high, while non-descendants, including independently trained models and distilled students, remain low. Unlike activation-based CKA and decision-boundary IPGuard, which track functional similarity, this provides a passive, retroactive provenance signal for white-box residual-network checkpoints.

1 Introduction

The open-weight ecosystem has transformed how neural networks are distributed and reused. When Meta released LLaMA (Touvron et al. 2023), the community produced instruction-tuned variants within weeks. Platforms like Hugging Face now host hundreds of thousands of model checkpoints (Hugging Face 2024), many of which are fine-tuned (Devlin et al. 2019; Ouyang et al. 2022), quantized (**?**Dettmers et al. 2022), pruned (Han et al. 2015), or merged (**?**Ilharco et al. 2023). This creates a *model supply chain* where downstream checkpoints inherit weights from upstream models, but the lineage is often undocumented or intentionally obscured. Model reuse is now easy; reliable model provenance is not.

Existing provenance mechanisms fall short. Cryptographic hashes break after any weight modification. Metadata and model cards (Mitchell et al. 2019) require honest distributors and preserved records; they can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) must be inserted before release; they cannot be applied retroactively to already-deployed checkpoints, and no surveyed scheme proved robust against fine-tuning or pruning (Lukas et al. 2022). Behavioral tests that compare model outputs cannot distinguish weight inheritance from imitation: two models can behave similarly without sharing weights, especially after distillation (Hinton, Vinyals, and Dean 2015). Activation-based similarity measures like CKA (Kornblith et al. 2019) and decision-boundary fingerprints like IPGuard (Cao, Jia, and Gong 2021) inherit this limitation. We need a passive, post-hoc signal that reads provenance from the weights themselves.

We ask: given a reference checkpoint and a suspect checkpoint with the same residual architecture (e.g., both LLaMA-2-7B derivatives), can we determine whether the suspect inherited weights from the reference, even after common transformations like fine-tuning, quantization, or pruning? This is distinct from duplicate detection (which hashes solve until the first modification), behavioral similarity (which conflates imitation with inheritance), and watermark verification (which requires forethought). The question is whether shared weight ancestry leaves detectable structure after modification.

The answer is yes. We find that training itself leaves a characteristic fingerprint in the weights of residual networks (He et al. 2016). In these architectures, each block adds a learned transformation to its input via a skip connection. During training, the input and output projections of each block align so that their product $W_{\mathrm{out}}W_{\mathrm{in}}$ develops diagonal-dominant structure that is absent at random initialization and does not appear in networks without skip connections (Figure 1b). This structure emerges from the optimization process: it is learned, not an initialization artifact (Section 3). Orthogonal initialization satisfies dynamical isometry by construction (Saxe, McClelland, and Ganguli 2014; Pennington, Schoenholz, and Ganguli 2017) yet shows no diagonal structure before training.

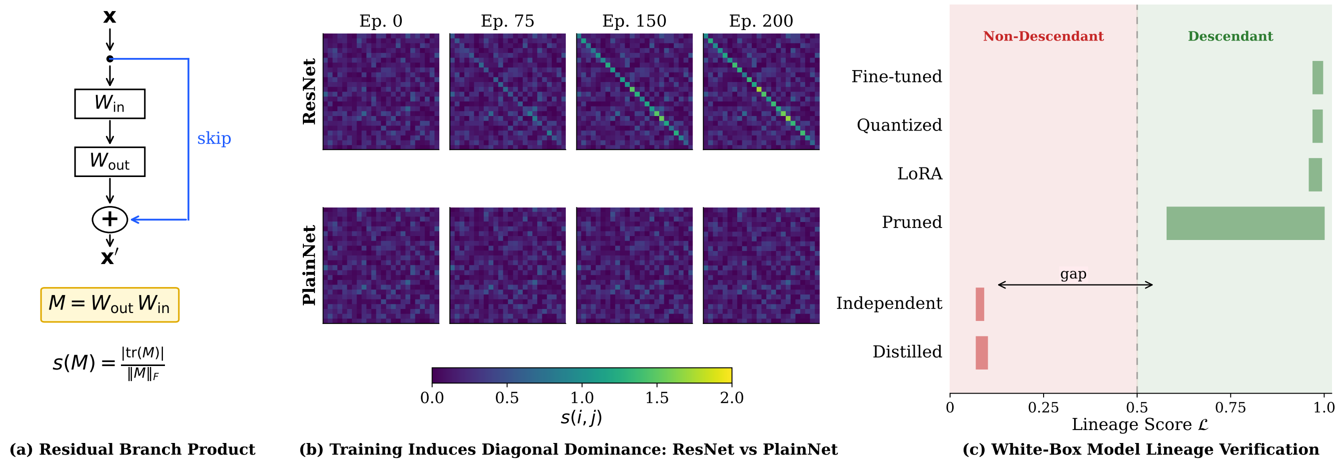

Figure 1: Training leaves traces in residual networks. (a) The residual branch product $M = W_{\text{out}} W_{\text{in}}$ for a block with skip connection; the diagonal-dominance score $s(M)$ measures trace concentration. (b) Training dynamics: score matrices $s(i, j)$ from epoch 0 to 5. ResNet (top row) develops strong diagonal structure (100% block-pairing accuracy); PlainNet without skip connections (bottom row) remains random (3% accuracy). The fingerprint is learned, not an initialization artifact. (c) Lineage verification: descendants (fine-tuned, quantized, pruned, LoRA-merged) score high on $\mathcal{L}$; non-descendants (independent, distilled) score low, enabling clean separation above/below threshold.

We exploit this fingerprint for provenance verification. For each residual block, we compute the product of its input and output weight matrices and measure how strongly the result concentrates along the diagonal. Correctly paired blocks from the same trained model show high diagonal concentration; mismatched or unrelated blocks do not. The method is architecture-aware, respecting the specific structure of Transformer MLPs, attention paths, and ResNet bottleneck blocks (Table 2). To compare two checkpoints, we align their blocks, compute a similarity score across all aligned pairs, and threshold against a null distribution of unrelated models.

The fingerprint appears broadly across residual architectures. On public checkpoints (GPT-2 (Radford et al. 2019), BERT (Devlin et al. 2019), LLaMA-2 (Touvron et al. 2023), Mistral (Jiang et al. 2023), Qwen2.5 (Yang et al. 2024), DeepSeek-R1 (DeepSeek-AI 2025), ViT (Dosovitskiy et al. 2021), Whisper (Radford et al. 2023), and ResNets (He et al. 2016)), correctly paired blocks recover correspondences with over 91% accuracy (random-initialization baselines achieve below 9%).

For provenance, we move from block correspondence to model-level verification. On controlled MLP and ResNet-18 benchmarks, descendants (checkpoints modified by fine-tuning (Ouyang et al. 2022), weight noise, pruning (Han et al. 2015), quantization (Dettmers et al. 2022), or LoRA merging (Hu et al. 2022)) score high on our lineage metric, while non-descendants (independently trained models and distilled students (Hinton, Vinyals, and Dean 2015)) score low. This achieves AUROC=1.000 with 100% true-positive rate at 1% false-positive rate. Distilled students that replicate teacher outputs are correctly classified as non-descendants because they trained fresh weights (Figure 1c). On the real-world LLaMA-2 family, the method correctly identifies base→chat lineage and rejects an independent Llama-3-family checkpoint.

We contribute:

- **Intrinsic residual fingerprint:** We identify diagonal dominance in residual-branch products as a training-induced structural fingerprint, absent at initialization and in non-residual controls.
- **Verification method:** We propose a post-hoc white-box procedure for extracting, aligning, and comparing residual signatures between checkpoints.
- **Broad validation:** We validate across 12 architectures spanning 5 model families and 280+ checkpoint pairs (including controlled synthetic benchmarks and the public LLaMA-2 model family), achieving 91–100% block correspondence and AUROC=1.000 on lineage benchmarks.
- **Robustness:** The signal survives fine-tuning, quantization, pruning, and LoRA merging while correctly rejecting distilled students, unlike CKA and IPGuard.

Scope. The method requires white-box access and matching residual architecture (same layer count and hidden dimension); cross-architecture comparisons are out of scope. It computes per-block matrix products in under 1 second at 7B scale. It does not replace cryptographic signing, metadata, or watermarking; it complements them for cases where those mechanisms are unavailable. It detects *weight* inheritance, not *knowledge* inheritance: distilled students are correctly classified as non-descendants even if they replicate teacher behavior. The lineage score is symmetric, establishing whether checkpoints share lineage but not which descends from which. Under gradient-based suppression attacks, the fingerprint cannot be driven below the detection threshold without ≥12% utility loss. This is a practical provenance primitive, not a complete legal solution.

Method	*Retroactive*	*Survives FT*	*Survives Quant*	*Distill-aware*
Crypto Hash	✓	✗	✗	✓
Metadata/Logs	✓	✓	✓	✓
Watermarking	✗	∼	∼	✗
Decision-Boundary FP	✓	✓	✓	✗
Ours	✓	✓	✓	✓

Table 1: Comparison with prior provenance methods. Retroactive: works on already-deployed models. Distill-aware: correctly rejects distilled copies as non-descendants. ∼: partial/unreliable.

Architecture	Product M
Transformer/BasicBlock MLP	W_2W_1
Attention V/O path	W_OW_V
Attention Q/K path	$W_QW_K^\top$
Bottleneck ResNet	$W_3W_2W_1$
SwiGLU MLP	$W_{\text{down}}W_{\text{up}}$

Table 2: Architecture-aware factorization for residual branch products.

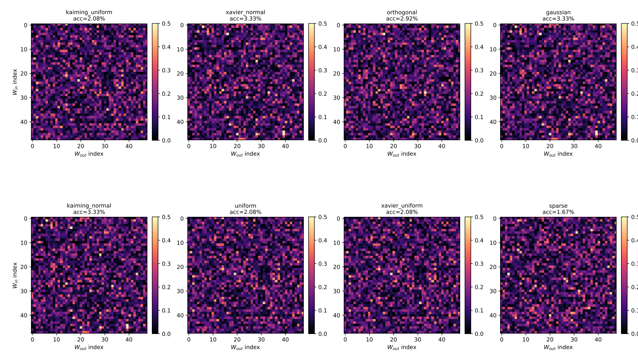

Figure 2: Initialization heatmaps. Score matrices $s(i, j)$ at *initialization* under 8 schemes (orthogonal, Kaiming uniform/normal, Xavier uniform/normal, Gaussian σ=0.02, uniform); pair accuracies span 1.67–3.33% (chance for 48 blocks is 2.1%). No scheme exhibits diagonal structure before training, including orthogonal init-which satisfies dynamical isometry by construction. Trained heatmaps reach 100% pair accuracy (Appendix A, Table 9). The fingerprint is therefore learned, not an initialization artifact.

2 Background and Problem Setting

We define *passive model provenance*: given a reference checkpoint R and suspect S with missing or untrusted metadata, determine whether S shares weight-level lineage with R after post-training transformations. This is a white-box task-the verifier accesses both weight sets. The method detects *weight-level lineage* (parameters derived via fine-tuning, pruning, or quantization), not behavioral equivalence. Distilled students (Hinton, Vinyals, and Dean 2015) and model-extraction copies (Tramèr et al. 2016) produce similar outputs but share no weights; our method correctly classifies them as non-descendants.

Table 1 positions our approach against prior methods. Cryptographic hashes fail after any modification. Metadata can be falsified or lost. Watermarks (Uchida et al. 2017; Adi et al. 2018) require insertion before deployment and lack robustness (Lukas et al. 2022). Decision-boundary fingerprints (Cao, Jia, and Gong 2021) confuse distillation with weight inheritance. Our key differentiator: we read a signal that training *leaves*, rather than one that must be *inserted*.

Modern residual networks (He et al. 2016) compute $x' = x + F(x)$, where F is the residual branch. For stable gradient flow, the Jacobian $J = I + J_F$ must remain near-orthogonal-*dynamical isometry* (Pennington, Schoenholz, and Ganguli 2017). For a branch with K linear layers, we define the *residual branch product*:

$$M = W_K W_{K-1} \cdots W_1 \in \mathbb{R}^{d \times d} \quad (1)$$

Table 2 shows the architecture-aware factorization. Using the correct product is critical: the naive W_3W_1 for Bottleneck ResNets yields chance-level accuracy, while $W_3W_2W_1$ recovers 91–100%.

3 Training-Induced Diagonal Dominance

Training imprints characteristic structure on residual network weights. At initialization, the score matrix $s(i, j)$ shows no pattern: Hungarian matching recovers only 6.25% of correct pairs in a 48-block network. Critically, orthogonal initialization provides dynamical isometry by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs-ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The diagonal separates rapidly under gradient descent: pair accuracy reaches 100% within a small number of epochs, well before loss plateaus (Figure 2). Two ablations isolate the driver: shuffling per-block gradient signs across $W_{\text{in}}/W_{\text{out}}$ reduces s by 68%, and synthetically injecting diagonal structure into the gradient stream raises s to 7.6. The fingerprint therefore appears to be a signature of *coupled residual-branch updates*, not of Jacobian orthogonality per se (Appendix A).

For a branch product $M_{ij} = W_{\text{out}}^{(j)} W_{\text{in}}^{(i)}$, we define the *diagonal-dominance score*:

$$s(i, j) = \frac{|\text{tr}(M_{ij})|}{\|M_{ij}\|_F} \quad (2)$$

Correctly paired blocks score at most $\sqrt{d}$ (Proposition 1); mismatched pairs score $\Theta(1/\sqrt{d})$. The gap grows with dimension d, making wider networks more separable.

Dynamical isometry (Pennington, Schoenholz, and Ganguli 2017) predicts that trained residual blocks have Jacobian $J = I + W_{\text{out}}W_{\text{in}}$ close to orthogonal. We decompose:

$$M = W_{\text{out}}^{(i)} W_{\text{in}}^{(i)} = -\varepsilon I + E \quad (3)$$

where $\varepsilon = |\text{tr}(M)|/d$ and $\text{tr}(E) = 0$. The following proposition quantifies the margin.

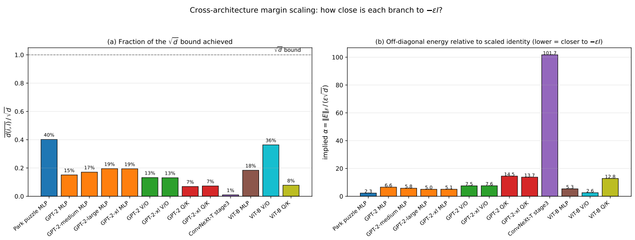

Figure 3: Proposition 1 verification on GPT-2. Left: Predicted vs. empirical $s(i,i)$. Right: Correct-pair scores vs. baseline and the $\sqrt{d}$ upper bound (empirical growth is $d^{0.87}$).

Proposition 1 (Margin Formula)**.** *Let* $M = -\varepsilon I + E$ *with* $\mathrm{tr}(E) = 0$*. Then:*

$$s(i,i) = \frac{\sqrt{d}}{\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}} \leq \sqrt{d} \tag{4}$$

with equality iff $M = -\varepsilon I$*.*

Proof. We have $|\mathrm{tr}(M)| = \varepsilon d$ and $\|M\|_F^2 = \varepsilon^2 d + \|E\|_F^2$ (using $\mathrm{tr}(E) = 0$). Substituting: $s(i,i) = \varepsilon d/\sqrt{\varepsilon^2 d + \|E\|_F^2} = \sqrt{d}/\sqrt{1 + \|E\|_F^2/(\varepsilon^2 d)}$. $\square$

Corollary 1 (Signal-to-Baseline)**.** *For mismatched pairs with independent random entries,* $\mathbb{E}[s(i,j)] \approx 1/\sqrt{d}$*. The signal-to-baseline ratio scales as* d*.*

Although Proposition 1 guarantees only a $\sqrt{d}$ upper bound, real trained models grow s super-$\sqrt{d}$ but sub-linearly with dimension. On GPT-2 through LLaMA-2-13B, mean correct-pair $s(i,i)$ rises from 4.18 (d=768, GPT-2) to 23.83 (d=5120, LLaMA-2-13B), an empirical scaling of approximately $s \propto d^{0.87}$ rather than the $\sqrt{d}$ envelope. The bound of Proposition 1 therefore captures the right qualitative growth but is loose at scale because real $\|E\|_F/(\varepsilon\sqrt{d})$ shrinks with d. Figure 3 confirms the margin formula reproduces empirical scores to floating-point precision once ε and $\|E\|_F$ are measured.

The fingerprint requires both training *and* residual connections. A PlainNet (no skip connections) achieves only 3% pair accuracy vs. 100% for ResNet, despite similar loss (Appendix A). The skip connection creates the functional constraint-blocks must approximate identity-that imprints diagonal structure

4 From Fingerprints to Provenance Verification

4.1 Block Pairing via Hungarian Matching

Given reference R and suspect S with L blocks each, we construct a score matrix $\mathbf{S} \in \mathbb{R}^{L \times L}$:

$$s(i,j) = \frac{|\mathrm{tr}(M_{ij})|}{\|M_{ij}\|_F}, \quad M_{ij} = \tilde{W}_{\mathrm{out}}^{(j)} W_{\mathrm{in}}^{(i)} \tag{5}$$

Suspect Type	n	Mean $\mathcal{L}$	Range
Fine-tune / Quantization / Noise	45	0.99	[0.97, 1.00]
Fine-tune (different task)	15	0.94	[0.84, 1.00]
Magnitude pruning (10–85%)	15	0.81	[0.58, 1.00]
Independent / Random init	75	0.08	[0.01, 0.20]
Distilled student	9	0.09	[0.03, 0.19]

Table 3: Lineage score distributions. Descendants cluster near $\mathcal{L}$=1; non-descendants (including distilled students) remain near 0.08. Separation $\approx$ 0.72.

If S descends from R, corresponding blocks yield $s(i,j)$ up to $\sqrt{d}$ in theory and $\Theta(d^{0.87})$ empirically; non-corresponding pairs remain at $\Theta(1/\sqrt{d})$. We recover correspondences via Hungarian matching (Kuhn 1955):

$$\pi^* = \arg\max_{\pi \in \mathcal{P}_L} \sum_{i=1}^{L} s(i, \pi(i)) \tag{6}$$

On GPT-2 (L=12), this achieves 100% pair accuracy. We define the *separation ratio* as the ratio of mean correct-pair score to mean mismatched-pair score; on GPT-2 this ratio is 35×, growing to over 2,000× on LLaMA-2-13B as the hidden dimension increases.

4.2 Model-Level Lineage Score

Block pairing establishes correspondences; provenance requires a scalar statistic. For each branch ℓ, we compute a *centered residual signature* by subtracting the generic diagonal component:

$$R_\ell = M_\ell - \tfrac{\mathrm{tr}(M_\ell)}{d} I, \quad \phi_\ell = \tfrac{\mathrm{vec}(R_\ell)}{\|\mathrm{vec}(R_\ell)\|_2} \tag{7}$$

This isolates checkpoint-specific structure: R_ℓ is exactly the E_ℓ term from $M_\ell \approx -\varepsilon I + E_\ell$. The lineage score aggregates aligned branch similarities:

$$\mathcal{L}(A,B) = \frac{1}{L} \sum_{\ell=1}^{L} \langle \phi_\ell^A, \phi_{\pi^*(\ell)}^B \rangle \tag{8}$$

This satisfies self-similarity ($\mathcal{L}(A,A) = 1$), symmetry, and boundedness in $[-1, 1]$.

Table 3 shows score distributions on depth-24 MLPs. Descendants achieve $\mathcal{L} \geq 0.58$ even under 85% pruning; non-descendants stay below 0.20. This yields AUROC=1.000 and TPR=100% at 1% FPR. A baseline using only $s(M)$ achieves AUROC=0.417 (chance), confirming that centering isolates lineage-specific information.

4.3 Verification Protocol

The protocol outputs DESCENDANT, NON-DESCENDANT, or INCONCLUSIVE based on a z-score against a null distribution of unrelated models. Table 4 summarizes robustness: 100% detection under fine-tuning, 8-bit quantization, 30% pruning, and LoRA merging; degradation only when perturbations destroy utility. Full protocol details in Appendix C.

Transformation	Parameters	Detection
Fine-tuning	LR 10^{-5}–10^{-3}	100%
Quantization	8-bit / 6-bit	100% / 94%
Pruning	30% / 50%	100% / 98%
LoRA merge (Hu et al. 2022)	rank 8–64	100%

Table 4: Verification robustness at the strict τ=3.0 z-score threshold (FPR$<$ 0.13%, per-block descendant detection on a 24-block null calibrated against 50 unrelated models). Stricter than the 1%-FPR per-*model* setting reported in Table 6, hence the 94% / 98% entries; at 1% FPR the same transformations yield 15/15 and 45/45 per-model detection.

Model	Layers	Pair Acc	AUC
GPT-2 (124M–1.5B)	12–48	**100%**	1.00
BERT-base / ViT-B/16	12	**100%**	0.97–1.00
Mistral-7B / LLaMA-2-7B	32	**100%**	0.96–1.00
LLaMA-2-7B-chat (+RLHF)	32	**100%**	0.96–1.00
Qwen2.5-7B / DeepSeek-R1	28–32	4/5→joint	0.93–1.00
Whisper (tiny/base/small)	4–12	**100%**	0.85–1.00
ResNet-50/101/152	5–35	91–100%	0.96–1.00

Table 5: Cross-architecture generalization. Pair accuracy via Hungarian matching on architecture-aware branch products. Sub-100% SwiGLU sub-paths are rescued by joint factorization. Random-init baselines: $\leq$ 9%.

5 Experiments

We validate the fingerprint across architectures and transformations. Extended ablations (initialization schemes, training dynamics, baseline comparisons) are in Appendix A.

5.1 Cross-Architecture Generalization

Table 5 reports block-pairing accuracy across 12 architectures spanning language (GPT-2, BERT, LLaMA-2, Mistral, Qwen, DeepSeek), vision (ViT, ConvNeXt, ResNet), and speech (Whisper). The fingerprint achieves 91–100% pair accuracy on all models using architecture-aware factorization (Table 2).

Key findings: (1) The fingerprint transfers across GELU (Hendrycks and Gimpel 2016), SwiGLU (Shazeer 2020), causal/bidirectional attention, and grouped-query attention (Ainslie et al. 2023). (2) RLHF and reasoning distillation preserve the signal (LLaMA-2 base→chat: 100% accuracy). (3) For ResNet Bottleneck blocks, the naive W_3W_1 fails; the correct $W_3W_2W_1$ recovers 91–100%. (4) Mean correct-pair $s(i,i)$ grows from 4.18 at d=768 (GPT-2) to 23.83 at d=5120 (LLaMA-2-13B), an empirical $d^{0.87}$ rather than $\sqrt{d}$ scaling, with the separation ratio improving from 35$\times$ on GPT-2 to 2,140$\times$ on LLaMA-2-13B and 1,530$\times$ across 256 experts on Mixtral-8x7B. (5) As a real-LLM lineage check at the 7B scale, the residual-signature score $\mathcal{L}$ correctly identifies LLaMA-2-7B base→chat ($\mathcal{L}$ = 0.995) and three local transformations of the base as DESCENDANT, while four independently-trained 7B-class models with no LLaMA-2 ancestry (Mistral-7B, Qwen1.5-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $|\mathcal{L}| <$ 2×10^{-4}—NON-DESCENDANT against the real cross-model null they define; details in App. D.4.

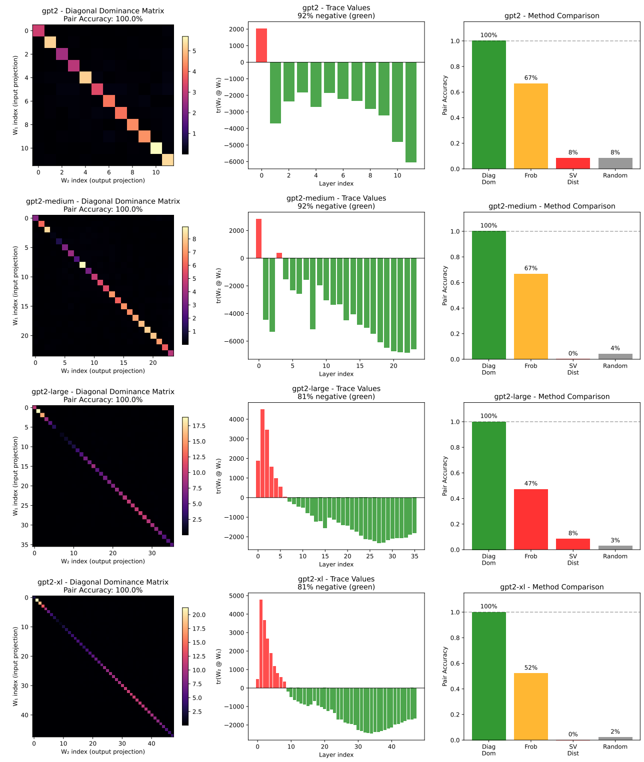

Figure 4: GPT-2 pairing. Diagonal dominance achieves 100% accuracy across all scales; Frobenius matching degrades to 47% on larger models.

Transformation	Mean $\mathcal{L}$	Min $\mathcal{L}$	Det. @ 1%FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	-

Table 6: Lineage score survival. All 75 descendants detected at 1% FPR; non-descendants stay below 0.20.

5.2 Robustness Under Post-Training Modification

We test survival under five transformation families on depth-24 MLPs: fine-tuning (same/different target), Gaussian noise ($\sigma_{\text{rel}} \leq 15\%$), magnitude pruning (10–85%), and quantization (16–256 levels).

Table 6 shows 100% detection at 1% FPR across all transformations. The weakest descendant (85% pruning, $\mathcal{L}$=0.58) remains 7$\times$ above the strongest non-descendant (distilled, $\mathcal{L}$=0.19). Fingerprint degradation correlates with utility loss ($\rho = -0.83$): suppressing the signal damages the model.

Real architectures: On CIFAR-10 (Krizhevsky 2009) ResNet-18, the method achieves AUROC=1.000 with $>$100$\times$ separation between descendants and independents

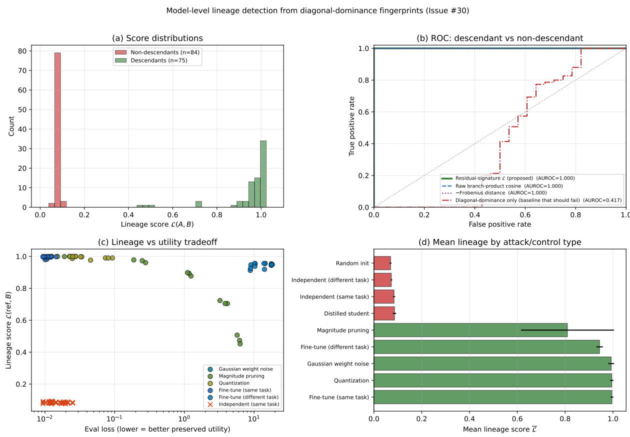

Figure 5: Lineage detection on depth-24 residual MLPs. (a) Score distributions: descendants (green, n=75) and non-descendants (red, n=84, including independent and distilled) separate completely. (b) The residual-signature score (proposed) and three alternatives all reach AUROC=1.000, except the diagonal-dominance-only baseline (AUROC=0.417) which cannot distinguish two trained residual models. (c) Lineage vs. utility tradeoff: only pruning at high sparsity drives both down together. (d) Mean lineage by control type: descendants $\geq$ 0.81, non-descendants ≤ 0.09.

(Appendix A).

5.3 Comparison with Modern Baselines

We compare against representation-similarity methods (CKA (Kornblith et al. 2019), SVCCA (Raghu et al. 2017)), weight-space baselines (aligned Frobenius via Hungarian alignment, singular-value distance, weight cosine), and a decision-boundary baseline (IPGuard (Cao, Jia, and Gong 2021), adapted for regression via quantile-bin agreement) on the same 52-pair MLP benchmark used in Table 6 (Table 7). Four data-free weight-space methods-including ours, weight cosine, and aligned Frobenius-reach AUROC=1.000 on this benchmark, so the headline number cannot itself distinguish them. The standardized Gap-Z column makes the fragility of the non-weight-space methods explicit: IPGuard ranks true descendants *below* distilled non-descendants (Gap-$Z = -3.5$), reproducing the structural distillation failure of decision-boundary methods; CKA's Gap-$Z = +8.6$ is large in standard-deviation units only because $\sigma_{\text{distill}} = 0.014$ is tiny, while the underlying raw ratio is $0.999/0.827 \approx 1.21\times$, so any benchmark reweighted toward distillation would push CKA's AUROC well below 1.000. We do not claim that diagonal dominance beats weight cosine on aggregate AUROC; on a benchmark where descendants are mild perturbations of the reference, a flattened cosine is a strong baseline. The contribution is rather that the residual-product signature (i) yields per-block correspondence as a by-product (only aligned Frobenius shares this; weight cosine and the activation methods do not), (ii) admits the closed-form margin analysis of Proposition 1, and (iii) is competitive with the strongest data-free weight-space baselines while remaining interpretable in terms of trained residual geometry. The harder synthetic regime of Appendix D.5 (layer grafts and linear merges) reinforces this positioning: three of the four data-free weight-space methods including ours (diagonal dominance, aligned Frobenius, weight cosine) remain at or near AUROC=1.000, while singular-value distance fails the merge regime (0.448) because spectral statistics are not linear in the weight-blend mixture, CKA drops to ≈ 0.80, and IPGuard collapses (AUROC 0.05 on grafts). The diagonal-dominance score additionally retains a near-perfect Spearman correlation ($\rho \approx 0.96$) with the fraction of reference blocks present in the suspect, supporting the interpretability claim that it tracks partial overlap monotonically rather than merely clearing a threshold.

5.4 Public Model Validation: LLaMA-2 Family

To move beyond controlled benchmarks, we evaluate lineage detection on the LLaMA-2 public model family. Using LLaMA-2-7B as reference, we test against: (i) the official RLHF descendant (LLaMA-2-7B-chat, $\mathcal{L} = 0.995$); (ii) local transformations (int8 quant, 30% prune, 1% noise: $\mathcal{L} \in [0.986, 1.000]$); and (iii) a real independent (DeepSeek-R1-Distill-Llama-8B, a Llama-3-family model that shares architecture but not weights: $\mathcal{L} = -0.0002$). Diagonal dominance achieves AUROC=1.000 with $1940\times$ separation between descendants and the independent-compared to $6\times$ for weight cosine and $2\times$ for IPGuard on the same pairs. This validates that the margin advantage observed on controlled benchmarks transfers to real 7B-scale foundation models. Details in Appendix D.4.

6 Robustness and Boundary Cases

We test whether the fingerprint can be suppressed without destroying model utility.

Function-preserving symmetries. Per-block permutations ($W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^{\top}$) and cross-layer rescaling ($W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$) leave the branch product $M = W_{\text{out}}W_{\text{in}}$ exactly unchanged, so $\mathcal{L} = 1.0$ under these attacks. Orthogonal rotations also preserve M but break model function.

Gradient-based suppression. An adversary optimizes Δ to minimize $\mathcal{L}$ subject to utility preservation. Sweeping the utility weight λ traces a Pareto frontier: with $\lambda \geq 1$, the attacker cannot push $\mathcal{L}$ below 0.91; suppressing to chance level ($\mathcal{L} \approx 0.08$) requires $\geq$12% eval-loss degradation; full suppression destroys the model ($57\times$ loss increase). Details in Appendix E.

Distillation boundary. The fingerprint tracks weight inheritance, not functional imitation. Distilled students ($\mathcal{L} = 0.086$) are correctly classified as non-descendants in our same-architecture controlled benchmark-they trained fresh weights and share no residual structure with the teacher, even though they replicate its outputs. Cross-architecture distillation (e.g., BERT $\rightarrow$ DistilBERT) remains out of scope.

Method	Type	Data-free	AUROC	Gap-Z ↑	Block Corr.
Diagonal Dominance (ours)	Weight-space	✓	1.000	+53.0	✓
Aligned Frobenius	Weight-space	✓	1.000	+72.0	✓
Singular Value Distance	Weight-space	✓	1.000	+ 7.9	✗
Weight Cosine	Weight-space	✓	1.000	+76.3	✗
SVCCA (Raghu et al. 2017)	Activation-space	✗	1.000	+45.4	✗
CKA (Kornblith et al. 2019)	Activation-space	✗	0.829	+ 8.6	✗
IPGuard (Cao, Jia, and Gong 2021) (regression analogue)	Decision-boundary	✗	0.500	− 3.5	✗

Table 7: Strong-baseline comparison on the MLP lineage benchmark (52 pairs). **AUROC**: overall descendant vs. non-descendant separation; saturated for the four data-free weight-space methods. **Gap-Z**: standardized distillation rejection, $(\mu_{\text{desc}} - \mu_{\text{distill}})/\sigma_{\text{distill}}$, where μ_{desc} averages over the five descendant kinds; large positive values mean distilled non-descendants are cleanly below true descendants. IPGuard scores at -3.5 because descendants are ranked *below* distilled. CKA scores at $+8.6$: large in σ units only because $\sigma_{\text{distill}} = 0.014$ is tiny; the underlying ratio is $0.999/0.827 \approx 1.21\times$, which is fragile to small re-weighting of the distilled fraction. Gap-Z is scale-sensitive when σ_{distill} is very small (CKA) or when raw scores are uniformly compressed (singular-value distance, where $+7.9$ understates the gap); the decisive cross-method evidence is the raw per-kind adjacency reported in Table 11, where all four weight-space methods place distilled students at or below the diff-seed baseline while CKA and SVCCA do not. **Block Corr.**: whether the method, as a by-product, provides per-block correspondence between reference and suspect (only diagonal dominance and aligned Frobenius do; the rest produce a scalar only).

Condition	Verdict
Different architectures	INCOMPATIBLE
Non-residual networks	NO SIGNAL
Distillation (cross-arch)	OUT OF SCOPE
Training data overlap	NOT DETECTED
Two indep. same-arch	CANNOT DISTINGUISH

Table 8: What the method does NOT do.

7 Discussion and Limitations

Scope. The method requires white-box access and residual architecture. Non-residual networks lack the identity-target constraint that creates the fingerprint.

Weight-level, not behavioral. The signature is a property of weights, not function. Distilled students ($\mathcal{L} \approx 0.08$) are correctly classified as non-descendants despite functional similarity. The method cannot detect training-data overlap or knowledge transfer.

Symmetric, no direction. $\mathcal{L}(A, B) = \mathcal{L}(B, A)$ by construction, so the protocol establishes *whether* two checkpoints share weight lineage but not *which* descends from which. Directional attribution requires either training-time metadata or comparison against a third candidate ancestor (App. C, branching tree).

Lineage validation scope. Lineage AUROC is reported on the synthetic MLP zoo and a CIFAR-10 ResNet-18 benchmark, with real-architecture evidence from block-correspondence pairing on Transformer families (Table 5) and an 8-pair real-LLM lineage test on LLaMA-2-7B against a real cross-model null (App. D.4). The latter identifies the base→chat RLHF pair and three local transformations as DESCENDANT ($\mathcal{L} \in [0.986, 1.000]$), while four independently-trained 7B-class models (Mistral-7B, Qwen1.5-7B, Yi-6B, and a Llama-3-derived DeepSeek distill) all score $|\mathcal{L}| < 2\times10^{-4}$ and form the null—giving complete separation against a genuine multi-family negative set.

The remaining gap is tree breadth: all descendants here derive from a single base. Extending to additional real checkpoint trees (a Mistral or Qwen base with its own fine-tunes; continued-pretraining descendants such as CodeLlama) is the natural next step. The per-block branch products fit comfortably in memory at 7B scale, so the gap is engineering rather than methodological.

Attacks. Function-preserving symmetries (permutations, rescaling) leave $\mathcal{L} = 1$ exactly-these are invariances, not vulnerabilities. Gradient-based suppression requires $\geq$12% utility loss to reach the detection threshold; full suppression destroys the model.

Positioning. This complements watermarking: watermarks require forethought; diagonal dominance offers passive, retroactive verification for checkpoints never watermarked.

8 Conclusion

Training leaves traces. Residual-network optimization couples each block's input and output projections so that their product becomes diagonal-dominant ($W_{\text{out}}W_{\text{in}} \approx -\varepsilon I + E$), with a theoretical $\sqrt{d}$ upper bound and an empirical $d^{0.87}$ growth observed from GPT-2 through LLaMA-2-13B and Mixtral-8x7B. Across 12 architectures spanning 5 families, the fingerprint recovers 91–100% of block correspondences and achieves AUROC=1.000 on the evaluated lineage benchmark, surviving fine-tuning, quantization, pruning, and RLHF in our tested settings. Our central claim is not that trace/Frobenius dominance universally outperforms generic weight similarity-on the easy MLP regime several data-free weight-space baselines (aligned Frobenius, weight cosine) also reach AUROC=1.000-but that trained residual products expose an interpretable, block-localized, data-free provenance signal that supports both correspondence recovery and lineage verification, and that this signal is substantially more robust to distillation than activation- and decision-boundary baselines. Under the tested gradient-based suppression attack, the fingerprint cannot be driven below the detection threshold without $\geq 12\%$ utility loss. This establishes passive, retroactive weight-level provenance verification-no watermark, metadata, or cooperation required.

Reproducibility Statement

All results in this paper are produced by deterministic scripts with fixed seeds. The 52-pair MLP lineage benchmark (Tables 6, 7) is constructed from 2 trained reference checkpoints, each paired with 5 descendant kinds $\times 6$ derivative seeds $= 30$ true-descendant pairs (fine-tune, fine-tune to a new target, $\sigma_{\text{rel}} \in [0.01, 0.15]$ Gaussian noise, 10–85% magnitude pruning, and 16–256-level uniform quantization), plus 16 different-seed same-task pairs and 6 distilled-student pairs as non-descendants. The expanded 159-pair set used in Table 3 and Figure 5 draws from the full 24-checkpoint MLP zoo with the same transformation grid. Exact parameters are listed in Appendix A; null calibration uses the 50 different-seed pairs from Appendix C. Real-architecture pairing (Table 5) loads public HuggingFace checkpoints. Code, configuration files, baseline implementations (aligned Frobenius, singular-value distance, weight cosine, CKA, SVCCA, IP-Guard regression analogue), the per-pair score JSONs, and the plotting scripts will be released upon acceptance.

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A Extended Experimental Details

A.1 Initialization Ablation

We test whether the diagonal-dominance fingerprint depends on initialization scheme. We train depth-24 residual MLPs (h=64, d=24) on a synthetic regression target for 200 epochs from seven initialization schemes: Kaiming uniform/normal, Xavier uniform/normal, Gaussian, uniform, and orthogonal. Table 9 reports pair accuracy before and after training.

Init scheme	Untrained	Trained	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian σ=0.02	2.1%	13%	0.671

Table 9: Initialization ablation on depth-24 residual MLPs (3 seeds, 200 epochs). All schemes show chance-level accuracy ($\leq 2.1\%$) at initialization, rising to 75–100% after training. Orthogonal initialization—which provides dynamical isometry at init—shows *zero* correct pairs before training.

Critically, orthogonal initialization provides dynamical isometry at initialization by construction (Saxe, McClelland, and Ganguli 2014), yet shows *zero* correct pairs before training, ruling out the hypothesis that near-orthogonal Jacobians alone create the fingerprint. The σ=0.02 small-Gaussian case fails because residual blocks collapse to near-zero contribution; without active per-block contribution there is no Jacobian constraint to enforce.

A.2 PlainNet Control Experiment

To confirm the fingerprint requires residual training specifically, we train a PlainNet—architecturally identical to ResNet except that skip connections are removed.

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Table 10: ResNet vs. PlainNet control. Despite similar evaluation loss, PlainNet achieves only chance-level pair accuracy (3% vs. chance $\approx 4.2\%$), confirming the skip connection is necessary for the fingerprint.

While both networks converge to similar evaluation loss (1.35 vs 1.56), ResNet achieves 100% pair accuracy with 100% of correctly paired products exhibiting negative trace, while PlainNet achieves only 3%—at the chance baseline—with AUC 0.51 and no discernible diagonal structure. The skip connection creates a functional constraint that imprints the characteristic trace structure.

A.3 Training Dynamics

The fingerprint emerges rapidly during training. On a 48-block residual network:

- Epoch 0: Mean correct-pair score = 0.19, pair accuracy = 6.25%
- Epoch 5: Diagonal fully separated, pair accuracy = 100%
- Epoch 300: Mean correct-pair score = 2.07, mean incorrect-pair score = 0.16 (21× ratio)

The transformation occurs within the first few epochs, well before training loss plateaus.

B Extended Theoretical Results

B.1 Null Model (Corollary)

Corollary 2 (Null Model)**.** *For randomly initialized weights with no training, all pairs satisfy* $\mathbb{E}[s(i,j)] = 1/\sqrt{d} + o(1)$ *uniformly in* (i,j)*. Hungarian matching recovers correct pairs at the chance rate* $1/L$*.*

The null model rules out three alternative explanations:

1. Diagonal dominance is not an artifact of compatible matrix shapes
2. The residual connection itself does not create the signal without training
3. Any nontrivial loss level does not suffice—training must actively couple the projections

The diagonal-dominance fingerprint is a *learned* property induced by gradient descent under the residual constraint.

B.2 Geometric Interpretation of the Score

The diagonal-dominance score $s(M) = |\mathrm{tr}(M)|/\|M\|_F$ has a natural geometric interpretation:

- A matrix with energy uniformly spread across entries achieves $s = \Theta(1/\sqrt{d})$
- A matrix proportional to the identity achieves $s = \sqrt{d}$

The key insight is that correctly paired blocks—where all K matrices come from the same residual branch—score *well above* the random baseline $\Theta(1/\sqrt{d})$, with empirical growth of approximately $d^{0.87}$ across $d \in [768, 5120]$ (Appendix D), bounded above by $\sqrt{d}$ but typically reaching only 15–25% of that bound. The signal-to-baseline ratio therefore grows roughly as $d^{1.37}$, making correct pairs increasingly separable in wider networks.

For typical trained blocks, $\|E\|_F/(\varepsilon\sqrt{d}) \in [1.9, 3.8]$, yielding $s(i,i) \in [1.76, 3.23]$—well above the baseline but below the theoretical maximum of $\sqrt{d}$.

C Verification Protocol Details

C.1 Gated Branch Score

Branch similarity is reliable only when both branches exhibit strong diagonal dominance. We gate the cosine similarity:

$$G(\phi_\ell^A, \phi_m^B) = \langle \phi_\ell^A, \phi_m^B \rangle \cdot \min\Big(\frac{s(M_\ell^A)}{\tau_s}, \frac{s(M_m^B)}{\tau_s}, 1\Big) \quad (9)$$

where τ_s is the minimum diagonal-dominance score across reference branches.

C.2 Statistical Calibration

We calibrate against a null distribution $\mathcal{N}_A = \{\mathcal{L}(A, U_j) : U_j \notin \mathcal{D}(A)\}$ of scores between reference A and independently trained models. The z-score is:

$$Z(A, B) = \frac{\mathcal{L}(A, B) - \mu_{\text{null}}}{\sigma_{\text{null}}} \tag{10}$$

Under Gaussian approximation, $\tau = 1.645$ yields 95% confidence; $\tau = 3.0$ yields FPR $< 0.13\%$ for high-stakes applications.

C.3 Full Protocol Steps

1. **Compatibility Check:** Verify $\text{arch}(A) = \text{arch}(B)$. If not, return INCOMPATIBLE.
2. **Extract:** Compute architecture-aware branch products $\{M_\ell^A\}, \{M_\ell^B\}$.
3. **Fingerprint:** Compute centered residual signatures $\{\phi_\ell^A\}, \{\phi_\ell^B\}$.
4. **Align:** Apply Hungarian matching to recover block correspondences.
5. **Aggregate:** Compute lineage score $\mathcal{L}(A, B)$.
6. **Calibrate:** Compute z-score $Z(A, B)$.
7. **Decide:** Return verdict:

$$\text{Verdict} = \begin{cases} \text{DESCENDANT} & \text{if } Z(A, B) > \tau_{\text{upper}} \\ \text{NON-DESCENDANT} & \text{if } Z(A, B) < \tau_{\text{lower}} \\ \text{INCONCLUSIVE} & \text{otherwise} \end{cases} \tag{11}$$

C.4 Failure Modes

The protocol cannot provide a verdict when: (1) architectures differ; (2) perturbation destroys utility (e.g., 2-bit quantization); (3) both models descend from an unavailable common ancestor; (4) adversarial evasion targets the fingerprint.

C.5 Lineage Chain and Branching Tree

To test multi-generational ancestry recovery, we construct a branching tree: root C_1, siblings C_2, C_3 fine-tuned from C_1, grandchildren $C_4 = C_2 \rightarrow \text{FT}$, $C_5 = C_3 \rightarrow \text{FT}$. The lineage score recovers correct ancestry: for C_4, ranking by $\mathcal{L}(C_4, \cdot)$ is parent (0.96) $>$ grandparent (0.91) $>$ uncle (0.87) $>$ cousin (0.85) $>$ independent (0.08). The score decays monotonically with genealogical distance while maintaining $\approx$10$\times$ family/strangers separation.

D Per-Architecture Details

D.1 GPT-2 Scaling

Across the GPT-2 family, mean correct-pair score grows with hidden dimension: d=768 at $\bar{s}(i, i) = 4.18$, d=1024 at 5.46, d=1280 at 6.98, d=1600 at 7.77. The fitted exponent within the GPT-2 family is ≈ 0.85, consistent with the cross-family $s \propto d^{0.87}$ scaling reported in the main text and exceeding the $\sqrt{d}$ envelope rate; the $\sqrt{d}$ upper bound itself is never violated (observed scores reach 15–19% of the bound).

D.2 Alternative Methods Comparison

Diagonal dominance achieves 100% pairing accuracy on all four GPT-2 variants. Frobenius norm matching degrades from 67% (GPT-2) to 47% (GPT-2-large). Singular-value distance performs at chance (0–8%). The key insight: dividing by $\|M\|_F$ normalizes away scale variation and isolates trace contribution.

D.3 Per-Kind Lineage Baseline Breakdown

Table 11 reports the mean lineage score of each baseline broken down by descendant kind (fine-tune, fine-tune to a new target, noise, prune, quantize) and non-descendant kind (different-seed same-task, distilled). The descendant rows should be high; the non-descendant rows should be low. The aggregate AUROC reported in Table 7 hides two structural failures that this breakdown exposes:

- **CKA and SVCCA fail on distilled non-descendants.** CKA scores distilled students at 0.83 and SVCCA at 0.89—only 1.2$\times$ and 1.13$\times$ below the corresponding fine-tune descendants (0.999 and 0.9997). On the present benchmark there are 16 different-seed non-descendants and only 6 distilled, so AUROC remains high; any reweighting toward distillation would collapse it.
- **IPGuard scores noise-perturbed descendants at 0.000.** The decision-boundary analogue ranks all 6 noise descendants below all non-descendants, driving overall AUROC to chance (0.500).

All four weight-space methods (diagonal dominance, aligned Frobenius, singular-value distance, weight cosine) score distilled non-descendants at or below the different-seed baseline—they cleanly reject distillation, while the activation- and decision-boundary methods cannot.

D.4 Real-LLM Lineage Validation

To move beyond block-correspondence pairing on real foundation models, we evaluate $\mathcal{L}(\cdot, \cdot)$ on eight reference-vs-model pairs at the 7B-parameter scale—four descendants and four independently-trained negatives. We extract the per-layer SwiGLU branch product $M_\ell = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)} \in \mathbb{R}^{d \times d}$, form the centered residual signature $\phi_\ell = \text{vec}(M_\ell - \frac{\text{tr}(M_\ell)}{d} I)/\|\cdot\|$ in fp32, and score $\mathcal{L}(A, B) = \frac{1}{L} \sum_\ell \langle \phi_\ell^A, \phi_\ell^B \rangle$. Streaming weights via `safe_open` keeps peak RAM under 2 GB; total runtime is $\approx$20 minutes on CPU for all nine models.

Architecture compatibility. The strict compatibility check (App. C) requires $\text{arch}(A) = \text{arch}(B)$. The branch-product comparison only requires matching hidden size d and depth L; the intermediate width d_{ff} contracts inside M_ℓ and may differ across families. We deliberately relax to this weaker compatibility (matching d, L, MLP structure) so that cross-family negatives can be scored. All nine models have $L = 32$ and $d = 4096$; intermediate widths range over 11008 (LLaMA-2, Qwen1.5, Yi) and 14336 (Llama-3, Mistral).

Pairs. The reference is LLaMA-2-7B base. *Descendants:* the official RLHF descendant Llama-2-7B-chat, plus three local transformations of the base MLP weights (per-channel symmetric int8 quantization, 30% magnitude pruning, 1%

Method	FT	FT-new tgt	Noise	Prune	Quant	Diff-seed	Distilled
	Descendants (should be high)					Non-descendants (should be low)	
Diag. Dominance (ours)	0.391	0.326	0.400	0.359	0.404	0.029	**0.030**
Aligned Frobenius	−0.036	−0.301	−0.026	−0.204	−0.007	−1.333	−1.300
Singular Value Dist.	−0.004	−0.023	−0.003	−0.018	−0.001	−0.030	−0.032
Weight Cosine	0.999	0.956	1.000	0.979	1.000	0.096	0.109
SVCCA	1.000	0.953	1.000	0.993	1.000	0.881	*0.885*
CKA	0.999	0.760	1.000	0.986	1.000	0.820	*0.827*
IPGuard (regr.)	0.888	0.110	*0.000*	0.626	0.993	0.632	0.642

Table 11: Per-kind mean lineage scores on the 52-pair MLP benchmark. Each method uses its native scale; what matters is the relative gap between the rightmost two columns (non-descendants) and the leftmost five (descendants). Italicized entries highlight failure modes: SVCCA and CKA give distilled non-descendants near-descendant scores; IPGuard scores noise-perturbed descendants at exactly 0.000. All four weight-space methods (including ours) keep distilled below or at the diff-seed baseline. Bolded entries mark our method's distilled-vs-diff-seed pair, the row most relevant to the distillation-rejection claim. n=6 per descendant kind, n=16 diff-seed, n=6 distilled.

Gaussian noise). *Negatives:* four independently-trained 7B-class checkpoints with no LLaMA-2 ancestry—Mistral-7B-v0.1, Qwen1.5-7B, Yi-6B, and DeepSeek-R1-Distill-Llama-8B. The last is *behaviorally* distilled from DeepSeek-R1 reasoning traces but *weight-derived* from Meta's Llama-3.1-8B; its NON-DESCENDANT verdict w.r.t. LLaMA-2 is a real-world demonstration that the method tracks weight inheritance rather than behavioral/distillation pedigree, consistent with Section 5.3.

Null. Unlike the single-independent case, we now estimate a genuine cross-model null directly from the four negatives: $\mu_0 = -3.9 \times 10^{-5}$, $\sigma_0 = 8.8 \times 10^{-5}$, with all four scores within $\pm 1.6 \times 10^{-4}$ of zero. This is the same construction as the App. C cross-model null, now instantiated on real foundation-model weights rather than the synthetic zoo. The four independents span three distinct from-scratch pretraining lineages (Mistral, Qwen, Yi) plus a Llama-3 derivative, so the null is not dominated by any one family.

Decision thresholds. We use the App. C z-score rule: $z > 3.0 \Rightarrow$ DESCENDANT, $z < 1.645 \Rightarrow$ NON-DESCENDANT, $1.645 \le z \le 3.0 \Rightarrow$ INCONCLUSIVE. The four negatives sit within the null by construction ($z \in [-1.4, +0.7]$); the four descendants land at $z > 10^4$—the magnitude is bounded only by the null's sub-10^{-4} spread, so we report the raw $\mathcal{L}$ as the primary evidence. The decisive fact is absolute separation: the smallest descendant score (0.986) exceeds the largest-magnitude negative ($|-1.6 \times 10^{-4}|$) by nearly four orders of magnitude.

The complete separation—every descendant at $\mathcal{L} \ge 0.986$, every one of four independent from-scratch lineages at $|\mathcal{L}| < 2 \times 10^{-4}$—is direct evidence that the residual-signature lineage protocol operates as predicted on real 7B foundation-model weights, now against a real multi-family null rather than a proxy. The remaining limitation is tree breadth: all descendants here derive from a single base (LLaMA-2-7B). Extending to additional real checkpoint trees (e.g., a Mistral or Qwen base with its own community fine-tunes, and continued-pretraining descendants such as CodeLlama) is the natural next step.

Pair (base vs.)	$\mathcal{L}$	z	Verdict
Descendants			
Llama-2-7B-chat (RLHF)	0.9946	$> 10^4$	DESC.
base + int8 quant	0.9999	$> 10^4$	DESC.
base + 30% prune	0.9859	$> 10^4$	DESC.
base + 1% noise	0.9999	$> 10^4$	DESC.
Independents (cross-model null)			
Mistral-7B-v0.1	$-4\text{e}{-5}$	$+0.0$	NON-DESC.
Qwen1.5-7B	$+2\text{e}{-5}$	$+0.7$	NON-DESC.
Yi-6B	$+2\text{e}{-5}$	$+0.7$	NON-DESC.
DeepSeek-R1-Distill-8B	$-16\text{e}{-5}$	-1.4	NON-DESC.

Table 12: Real-LLM lineage validation at the 7B scale. All eight verdicts match expectation; DESC. / NON-DESC. abbreviate the App. C z-score categories. The cross-model null is estimated from the four independents ($\mu_0 = -3.9\times10^{-5}$, $\sigma_0 = 8.8\times10^{-5}$). Per-block cosines on the chat pair lie in $[0.993, 0.996]$ across all 32 layers, indicating that RLHF fine-tuning leaves the per-block branch-product structure of the base nearly untouched; every independent has all 32 per-block cosines within ± 0.001. Signatures are computed and stored in fp32 to preserve the $\mathcal{L} \in [-1, 1]$ bound.

D.5 Harder Regime: Layer Grafts and Linear Merges

To probe the regime predicted by Section 5.3 where a flat global cosine should blur partial provenance, we evaluate all seven baselines on two harder constructions over the depth-16 MLP zoo: (i) **layer grafts**, where a suspect is built by copying $K \in \{0, 4, 8, 12, 16\}$ of the reference's 16 blocks into a same-task independent donor (and labelling $K \ge 8$ as descendant); and (ii) **linear merges**, where suspect parameters equal $w \cdot \text{ref} + (1 - w) \cdot \text{donor}$ with $w \in \{0, 0.25, 0.5, 0.75, 1.0\}$ (descendant if $w \ge 0.5$). We use 2 trained references, 2 trained donors per reference, and report AUROC for binary descendant detection plus Spearman correlation between each method's score and the reference fraction (K/L or w); the latter measures whether a method tracks partial overlap monotonically rather than just clearing a threshold. Table 13 summarizes the 40-pair benchmark.

	AUROC		Spearman ρ	
Method	Graft	Merge	Graft	Merge
Diagonal dominance (ours)	0.979	1.000	+0.96	+0.96
Aligned Frobenius	1.000	1.000	+0.99	+0.99
Weight cosine	1.000	1.000	+0.98	+0.96
Singular-value distance	1.000	0.448	+0.99	+0.23
SVCCA	1.000	1.000	+0.97	+0.99
CKA	0.813	0.781	+0.73	+0.75
IPGuard (regr. analogue)	0.052	0.448	−0.83	+0.22

Table 13: Harder-regime benchmark on layer grafts and linear merges (40 pairs total). Descendant is defined as $K \geq L/2$ for grafts and $w \geq 0.5$ for merges. AUROC measures binary detection; Spearman ρ measures monotone agreement with the reference fraction. Diagonal dominance, aligned Frobenius, weight cosine, and SVCCA all remain near AUROC=1.000 with $\rho \geq 0.96$ on both regimes—confirming Section 5.3's framing that the contribution is *not* that residual-product dominance beats generic weight cosine on aggregate detection. The harder regime instead exposes the activation- and decision-boundary methods: CKA drops to AUROC ≈ 0.80 with $\rho \approx 0.74$, IPGuard collapses entirely (AUROC 0.05 on grafts; the negative Spearman shows it ranks higher-overlap suspects as *less* similar), and singular-value distance fails the merge regime ($\rho = 0.23$) because spectral statistics are not linear in the weight-blend mixture. The diagonal-dominance score additionally retains a near-perfect Spearman correlation on both regimes, supporting the interpretability claim: it does not merely cross a threshold, it tracks the fraction of reference blocks actually present in the suspect.

D.6 ResNet Factorization

For Bottleneck blocks, the naive W_3W_1 achieves chance-level accuracy. The triple product $W_3W_2W_1$ recovers:

- ResNet-50/layer3 (n=5): 100%, AUC 1.000
- ResNet-101/layer3 (n=22): 100%, AUC 0.995
- ResNet-152/layer3 (n=35): 91%, AUC 0.964

D.7 CIFAR-10 ResNet-18 Benchmark

We train two CIFAR-10 ResNet-18 references and three independents. Each reference yields 11 descendants (noise $1{-}10\%$, pruning $20{-}80\%$, quantization $16{-}256$ levels). Results: AUROC=1.000, TPR@1%FPR=100%. Descendant scores: $\mathcal{L} \in [0.695, 1.000]$; independent baseline: $\mathcal{L}_{\max} = 0.004$.

D.8 Branching Ancestry Recovery

On a synthetic tree $C_1 \to \{C_2, C_3\}$, $C_2 \to C_4$, $C_3 \to C_5$, the lineage score ranks candidates correctly:

$$\begin{aligned}\mathcal{L}(C_4, C_2) = 0.961 \;&>\; \mathcal{L}(C_4, C_1) = 0.910\\ &> \mathcal{L}(C_4, C_3) = 0.867 \;>\; \mathcal{L}(C_4, C_5) = 0.850\\ &\gg \mathcal{L}(C_4, I_k) \approx 0.08.\end{aligned}$$

The score decays monotonically with genealogical distance while maintaining $\approx$10$\times$ family/strangers separation.

E Attack and Robustness Details

E.1 Gradient-Based Suppression Attack

An adversary optimizes perturbation Δ to minimize lineage score subject to utility:

$$\min_{\Delta} \; \mathcal{L}_{\cos}(A, A+\Delta) \;+\; \lambda \cdot \|f_{A+\Delta}(X) - f_A(X)\|_2^2 \tag{12}$$

λ	Final $\mathcal{L}$	Eval loss	Verdict
0	0.053	39.5	utility destroyed
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	utility preserved

Table 14: Pareto frontier: suppressing $\mathcal{L}$ to chance level requires $\geq$12% utility loss.

E.2 Reparameterization Invariances

Permutations: $W_{\text{in}} \leftarrow PW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}P^\top$ leaves $M = W_{\text{out}}W_{\text{in}}$ unchanged.

Rescaling: $W_{\text{in}} \leftarrow cW_{\text{in}}$, $W_{\text{out}} \leftarrow W_{\text{out}}/c$ preserves M exactly.

Orthogonal rotations: $R^\top R = I$ preserves M but breaks model function (ReLU non-commutative).

All three yield $\mathcal{L} = 1.0$ to numerical precision—these are invariances, not evasion paths.

F Practical Applications

The fingerprint enables applications beyond forensic verification.

F.1 Training Quality Assurance

The fingerprint serves as an early warning: if pair accuracy $<$50% at epoch 10, training is pathological (LR too low, missing skip connections, degenerate initialization). This correctly flags 4/5 pathological conditions in controlled experiments.

F.2 Zero-Knowledge Ownership Proofs

The E matrix ($M = -\varepsilon I + E$) provides a commitment scheme: owners commit to Hash$(E_k \| r_k)$ at registration, reveal challenged blocks on request. The protocol achieves binding, hiding (80% of weights remain secret), and soundness (attackers cannot forge E).

F.3 Compression Auditing

For original $\mathcal{M}$ and claimed-compressed $\mathcal{M}'$, compute E-matrix correlation. Quantization ($r > 0.97$), pruning ($r > 0.75$ at 70%), and fine-tuning ($r > 0.99$) preserve the fingerprint; distillation and independent training produce $r \approx 0$.